Site Analysis: From Environmental Evidence to Design Response

ARCHITECTURE · architecture-architectural-design-and-site-planning · BoardPrep lesson v2

Official examination scope

The PRC June 2026 Architects program assigns 40% to Architectural Design and Site Planning for that cycle; it does not weight this lesson topic. This lesson combines original instruction with MIT course references. Confirm future program changes and applicable Philippine accessibility, safety, and building rules.

Learning objectives

- Separate measured site conditions, observations, and assumptions before choosing a design
- response.
- Compare orientation, access, adjacency, circulation, and site capacity against a stated user
- or program need.
- Explain a design choice with evidence, trade-offs, and the checks still required for
- compliance.

1. Map relationships before placing functions

Begin with the north arrow, scale, contours or spot elevations, arrival points, neighboring uses, vegetation, drainage, and stated hazards. Mark which observations are measured, supplied, or assumed. A circulation sketch shows how people and service access move; an adjacency sketch shows which functions need to connect or stay apart. Keep the evidence visible beside the proposed layout so a reviewer can trace each choice back to a site fact. A diagram does not establish soil stability, drainage performance, accessible grades, or code compliance.

2. Translate evidence into comparisons

For each proposed use, compare when sunlight reaches an edge, shade, usable daylight, glare, ventilation, view, noise, and privacy. Use the time, season, and climate details supplied in the problem; compass direction alone is not a universal answer. Test orientation, openings, and shading together because reducing direct sun can also reduce daylight or views. State which condition the proposed response improves and what still needs measurement or simulation.

3. Resolve trade-offs explicitly

Compare a small number of layout options against program fit, usable area, public and service circulation, accessible route continuity, adjacency, grade, drainage, hazards, and any stated site limits. A flat location may still be unsafe or poorly connected. Name the constraint each option addresses and the trade-off it creates. Check zoning, setbacks, fire safety, accessibility, utilities, and structural requirements against the applicable local rules and qualified professionals; this lesson does not determine compliance.

4. Communicate a reasoned proposal

Use a decision chain: supplied evidence, user or program need, design response, then verification. Label assumptions and show circulation, environmental conditions, and functional relationships in diagrams that answer the actual question. Compare options using explicit criteria instead of unlabeled preference. If the scenario lacks information needed for a final choice, state which fact would change the decision rather than claiming the proposal is universally best.

Place a reading room from a site schedule

Question: The west edge has late-afternoon direct sun, the north edge is shaded, and pedestrians arrive from the east. A reading room needs usable daylight with limited direct glare.
1. Map the east arrival path and keep the entrance connection legible.
2. Compare a north-side room with west exposure and the daylight the room still needs.
3. Test shading, glazing, views, and circulation together.
Answer: Test a north-side location and shading against daylight, glare, and east-side arrival.
Check: The proposal follows the supplied schedule and is not a universal orientation rule.

Compare two access routes on a sloping site

Question: The entrance is on the east side. One route follows the contour with a longer path; a shorter route crosses the slope directly. What should the designer compare?
1. Mark the entrance, destinations, contour direction, drainage path, and any known hazards.
2. Compare route continuity, grade changes, landings, visibility, drainage, and distance for both options.
3. Check the applicable accessibility requirements and site measurements before calling either route compliant.
Answer: Choose the route that best supports a continuous, legible, usable connection after the required measurements and code checks.
Check: The shorter route is not automatically better, and following a contour is not proof of compliance.

Common misconceptions

- Treating compass direction as a complete design solution without climate or user evidence.
- Placing the entrance at geometric center while ignoring actual arrival paths.
- Choosing a flat location without checking safety, drainage, access, and use.
- Calling a diagram complete after mapping only the boundary and north arrow.

Practice patterns

- For site scenarios, underline the supplied site facts and user need before selecting a
- response.
- For orientation questions, compare sun, shade, access, and use together; do not apply a fixed
- compass rule.
- For route or zoning questions, check safety, accessibility, adjacency, drainage, and program
- trade-offs.

Official scope references

Official scope: Program for the June 8 and 10, 2026 Licensure Examination for Architects — Professional Regulation Commission; Professional Regulatory Board of Architecture — https://www.prc.gov.ph/sites/default/files/Program%20for%20June%202026%20LEA%28Architect%29_rev.pdf

Instructional references

Instructional reference: Introduction to Architecture and Environmental Design — MIT OpenCourseWare, Department of Architecture — https://ocw.mit.edu/courses/4-111-introduction-to-architecture-environmental-design-spring-2014/
Instructional reference: Environmental Technologies in Buildings — MIT OpenCourseWare, Department of Architecture — https://ocw.mit.edu/courses/4-401-environmental-technologies-in-buildings-fall-2018/

BoardPrep lesson material is original instruction. It is not an official examination document.
No claim about question frequency or examination weighting is made.